BAILEY W. ROCKHOLT

Environmental GIS and Spatial Data Analysis

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PROFESSIONAL SUMMARY

Environmental GIS professional with experience in California State Parks program support, USDA geospatial AI research, and scientific mapping. Use ArcGIS Pro, ArcGIS Online, R, and Python to prepare data, analyze environmental patterns, and communicate results. Combine an MBA and science background with teaching, technical documentation, and project coordination experience.

TECHNICAL SKILLS

GIS and cartography - ArcGIS Pro, ArcGIS Online, QGIS, Experience Builder, Instant Apps, ModelBuilder, map layouts, symbology, and pop-ups.

Data and analysis - R, Python, basic SQL, Excel, geodatabases, hosted feature layers, spatial and tabular joins, projections, data QA/QC, terrain analysis, and remote sensing.

Field and research workflows - Survey123, Field Maps, QuickCapture, GPS data, environmental mapping, technical documentation, and stakeholder communication.

RELEVANT EXPERIENCE

Parks Specialist, Grants and Loans

California State Parks, Division of Boating and Waterways | August 2026 - Present

- Create and update project maps and spatial summaries for boating access, shoreline, vessel-removal, and environmental infrastructure programs.
- Connect project locations, grant records, imagery, and attributes; review spatial and tabular information for accuracy and consistency.
- Review agreements, amendments, budgets, and reimbursements; track funding and deadlines, maintain program records, and coordinate with agencies and fiscal staff.

Geospatial AI Intern

USDA National Agricultural Statistics Service | Remote | August 2026 - Present

- Support exploratory agricultural GeoAI workflows using aerial and satellite imagery, land-use layers, Python, and spatial datasets.
- Investigate imagery-based feature extraction and object detection; document prototype methods and outputs for research evaluation.

Applied GIS and Web Mapping

Cuesta College and independent portfolio | January 2026 - Present

- Prepare geodatabases, shapefiles, hosted layers, and attribute tables; perform joins, projection review, spatial queries, and data-quality checks.
- Build web maps and applications with symbology and pop-ups; develop mobile collection workflows and document repeatable GIS processes.

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SELECTED PROJECTS AND RESEARCH

California Wildfire Spatial Analysis | R and ArcGIS Online - Cleaned incident records, summarized annual and seasonal patterns and county-level fire frequency and acres burned, and communicated results through charts and a public web map.

Lake Tahoe Wildfire and Land Cover | ArcGIS Pro - Combined wildfire perimeters, NLCD land cover, elevation, roads, and water features; organized raster and vector layers and produced a cartographic layout.

Clarkia Habitat and Climate | Botanical research support - Mapped study locations and climate variables to support a student thesis on plant drought resistance. Used elevation, precipitation, temperature, labels, and symbology to show site context.

Colorado Alpine Pond Research | Ongoing GIS contribution - Prepare study-area and terrain maps using GPS/sample locations and USGS elevation data; develop and refine cartographic layouts for scientific communication.

California Crop Suitability | Python prototype - Built a command-line tool that ranks crops using weighted environmental scores, county presets or custom inputs, factor-level explanations, and CSV export. Uses illustrative assumptions rather than a validated agronomic model.

EDUCATION

Geographic Information Systems Certificate
Cuesta College | Expected December 2026

Coursework includes GIS, web applications, remote sensing, advanced GIS, and sustainable agriculture.

Master of Business Administration
University of California, Riverside | June 2025

B.S. in Nutrition and Food, Cum Laude
California State University, Sacramento | December 2022

ADDITIONAL PROFESSIONAL EXPERIENCE

Head Cross Country and Track Coach / Kinesiology Instructor
Porterville College | December 2025 - Present

- Teach kinesiology, assess student work, and maintain course records; coordinate training, travel, equipment, and meet logistics, using maps for route visualization.

Director of Operations / Assistant Coach
Claremont-Mudd-Scripps Colleges | July 2025 - August 2026

- Managed an operating budget of approximately $30,000 per month and coordinated 18 team trips; maintained expense records and worked with vendors and campus partners.

Teaching Assistant and Curriculum Developer
University of California, Riverside | December 2023 - June 2025

- Led business discussions and biology laboratories for 20-50 students; supported data analysis and scientific writing, and developed nutrition teaching materials and an ArcGIS Online learning activity.

ENVIRONMENTAL FIELD EXPERIENCE

Supported Santa Ana sucker research through fish handling, spawning support, and data collection; contributed to Fisheries Resource Volunteer Corps cleanups and outreach; participated in bighorn sheep monitoring and aquaponics work.